

by_normal_form_q13 over GF(13)

June 30, 2026

The equation

The equation of the quartic curve is :

$$X_0^4 + X_1^4 + 4X_2^4 + 7X_0^3X_1 + 5X_0^3X_2 + 5X_0X_1^3 + X_1^3X_2 + 7X_0X_2^3 + 2X_1X_2^3 + 3X_0^2X_1^2 + 11X_0^2X_2^2 + X_1^2X_2^2 + 8X_0^2X_1X_2 + 7X_0X_1^2X_2 + X_0X_1X_2^2$$

$$(1, 1, 4, 7, 5, 5, 1, 7, 2, 3, 11, 1, 8, 7, 1)$$

$$X_0^4 + X_1^4 + 4X_2^4 + 7X_0^3X_1 + 5X_0^3X_2 + 5X_0X_1^3 + X_1^3X_2 + 7X_0X_2^3 + 2X_1X_2^3 +$$

General information

Number of bitangents	28
Number of points	8
Fullness	is full
Number of Kovalevski points	9
Bitangent line type (a_0, a_1, a_2)	(24, 0, 4)
Number of singular points	0

All points on the curve

The quartic has 8 points:

The points on the quartic curve are:

$$\begin{array}{lll} 0 : P_{35} = (8, 1, 1) & 3 : P_{84} = (5, 5, 1) & 6 : P_{176} = (6, 12, 1) \\ 1 : P_{39} = (12, 1, 1) & 4 : P_{110} = (5, 7, 1) & 7 : P_{178} = (8, 12, 1) \\ 2 : P_{80} = (1, 5, 1) & 5 : P_{112} = (7, 7, 1) & \end{array}$$

The points by rank are: (35, 39, 80, 84, 110, 112, 176, 178)

The Kovalevski points are:

$$0 : P_{113} = (8, 7, 1) = a_6 \cap b_1 \cap c_{16} \cap d$$

1 : $P_{109} = (4, 7, 1) = a_2 \cap b_6 \cap c_{26} \cap d$
 2 : $P_{53} = (0, 3, 1) = a_3 \cap a_4 \cap c_{23} \cap c_{24}$
 3 : $P_{11} = (8, 1, 0) = a_5 \cap a_6 \cap c_{25} \cap c_{26}$
 4 : $P_5 = (2, 1, 0) = c_{13} \cap c_{14} \cap c_{35} \cap c_{45}$
 5 : $P_{179} = (9, 12, 1) = a_6 \cap b_2 \cap c_{15} \cap c_{34}$
 6 : $P_{175} = (5, 12, 1) = b_2 \cap b_5 \cap c_{26} \cap c_{56}$
 7 : $P_4 = (1, 1, 0) = b_3 \cap b_4 \cap c_{36} \cap c_{46}$
 8 : $P_0 = (1, 0, 0) = a_1 \cap b_2 \cap c_{12} \cap d$

The incidence structure induced by the Kovalevski points is:

Flags=(8, 10, 20, 29, 39, 45, 48, 50, 54, 68, 69, 71, 79, 88, 96, 100, 116, 121, 130, 140, 144, 155, 164, 174, 181, 183, 186, 194, 202, 214, 220, 232, 240, 243, 244, 251)

nb_rows=28

nb_cols=9

nb_flags=36

Incma=

(0, 0, 0, 0, 0, 0, 0, 0, 1)
 (0, 1, 0, 0, 0, 0, 0, 0, 0)
 (0, 0, 1, 0, 0, 0, 0, 0, 0)
 (0, 0, 1, 0, 0, 0, 0, 0, 0)
 (0, 0, 0, 1, 0, 0, 0, 0, 0)
 (1, 0, 0, 1, 0, 1, 0, 0, 0)
 (1, 0, 0, 0, 0, 0, 0, 0, 0)
 (0, 0, 0, 0, 0, 1, 1, 0, 1)
 (0, 0, 0, 0, 0, 0, 0, 1, 0)
 (0, 0, 0, 0, 0, 0, 0, 1, 0)
 (0, 0, 0, 0, 0, 0, 1, 0, 0)
 (0, 1, 0, 0, 0, 0, 0, 0, 0)
 (0, 0, 0, 0, 0, 0, 0, 0, 1)
 (0, 0, 0, 0, 1, 0, 0, 0, 0)
 (0, 0, 0, 0, 1, 0, 0, 0, 0)
 (0, 0, 0, 0, 0, 1, 0, 0, 0)
 (1, 0, 0, 0, 0, 0, 0, 0, 0)
 (0, 0, 1, 0, 0, 0, 0, 0, 0)
 (0, 0, 1, 0, 0, 0, 0, 0, 0)
 (0, 0, 0, 1, 0, 0, 0, 0, 0)
 (0, 1, 0, 1, 0, 0, 1, 0, 0)
 (0, 0, 0, 0, 0, 1, 0, 0, 0)
 (0, 0, 0, 0, 1, 0, 0, 0, 0)
 (0, 0, 0, 0, 0, 0, 0, 1, 0)
 (0, 0, 0, 0, 1, 0, 0, 0, 0)
 (0, 0, 0, 0, 0, 0, 0, 1, 0)
 (0, 0, 0, 0, 0, 0, 1, 0, 0)
 (1, 1, 0, 0, 0, 0, 0, 0, 1)

The Kovalevski points sorted by rank are: (0, 4, 5, 11, 53, 109, 113, 175, 179)

The points off the curve are:

0 : $P_0 = (1, 0, 0)$	44 : $P_{46} = (6, 2, 1)$	88 : $P_{92} = (0, 6, 1)$
1 : $P_1 = (0, 1, 0)$	45 : $P_{47} = (7, 2, 1)$	89 : $P_{93} = (1, 6, 1)$
2 : $P_2 = (0, 0, 1)$	46 : $P_{48} = (8, 2, 1)$	90 : $P_{94} = (2, 6, 1)$
3 : $P_3 = (1, 1, 1)$	47 : $P_{49} = (9, 2, 1)$	91 : $P_{95} = (3, 6, 1)$
4 : $P_4 = (1, 1, 0)$	48 : $P_{50} = (10, 2, 1)$	92 : $P_{96} = (4, 6, 1)$
5 : $P_5 = (2, 1, 0)$	49 : $P_{51} = (11, 2, 1)$	93 : $P_{97} = (5, 6, 1)$
6 : $P_6 = (3, 1, 0)$	50 : $P_{52} = (12, 2, 1)$	94 : $P_{98} = (6, 6, 1)$
7 : $P_7 = (4, 1, 0)$	51 : $P_{53} = (0, 3, 1)$	95 : $P_{99} = (7, 6, 1)$
8 : $P_8 = (5, 1, 0)$	52 : $P_{54} = (1, 3, 1)$	96 : $P_{100} = (8, 6, 1)$
9 : $P_9 = (6, 1, 0)$	53 : $P_{55} = (2, 3, 1)$	97 : $P_{101} = (9, 6, 1)$
10 : $P_{10} = (7, 1, 0)$	54 : $P_{56} = (3, 3, 1)$	98 : $P_{102} = (10, 6, 1)$
11 : $P_{11} = (8, 1, 0)$	55 : $P_{57} = (4, 3, 1)$	99 : $P_{103} = (11, 6, 1)$
12 : $P_{12} = (9, 1, 0)$	56 : $P_{58} = (5, 3, 1)$	100 : $P_{104} = (12, 6, 1)$
13 : $P_{13} = (10, 1, 0)$	57 : $P_{59} = (6, 3, 1)$	101 : $P_{105} = (0, 7, 1)$
14 : $P_{14} = (11, 1, 0)$	58 : $P_{60} = (7, 3, 1)$	102 : $P_{106} = (1, 7, 1)$
15 : $P_{15} = (12, 1, 0)$	59 : $P_{61} = (8, 3, 1)$	103 : $P_{107} = (2, 7, 1)$
16 : $P_{16} = (1, 0, 1)$	60 : $P_{62} = (9, 3, 1)$	104 : $P_{108} = (3, 7, 1)$
17 : $P_{17} = (2, 0, 1)$	61 : $P_{63} = (10, 3, 1)$	105 : $P_{109} = (4, 7, 1)$
18 : $P_{18} = (3, 0, 1)$	62 : $P_{64} = (11, 3, 1)$	106 : $P_{111} = (6, 7, 1)$
19 : $P_{19} = (4, 0, 1)$	63 : $P_{65} = (12, 3, 1)$	107 : $P_{113} = (8, 7, 1)$
20 : $P_{20} = (5, 0, 1)$	64 : $P_{66} = (0, 4, 1)$	108 : $P_{114} = (9, 7, 1)$
21 : $P_{21} = (6, 0, 1)$	65 : $P_{67} = (1, 4, 1)$	109 : $P_{115} = (10, 7, 1)$
22 : $P_{22} = (7, 0, 1)$	66 : $P_{68} = (2, 4, 1)$	110 : $P_{116} = (11, 7, 1)$
23 : $P_{23} = (8, 0, 1)$	67 : $P_{69} = (3, 4, 1)$	111 : $P_{117} = (12, 7, 1)$
24 : $P_{24} = (9, 0, 1)$	68 : $P_{70} = (4, 4, 1)$	112 : $P_{118} = (0, 8, 1)$
25 : $P_{25} = (10, 0, 1)$	69 : $P_{71} = (5, 4, 1)$	113 : $P_{119} = (1, 8, 1)$
26 : $P_{26} = (11, 0, 1)$	70 : $P_{72} = (6, 4, 1)$	114 : $P_{120} = (2, 8, 1)$
27 : $P_{27} = (12, 0, 1)$	71 : $P_{73} = (7, 4, 1)$	115 : $P_{121} = (3, 8, 1)$
28 : $P_{28} = (0, 1, 1)$	72 : $P_{74} = (8, 4, 1)$	116 : $P_{122} = (4, 8, 1)$
29 : $P_{29} = (2, 1, 1)$	73 : $P_{75} = (9, 4, 1)$	117 : $P_{123} = (5, 8, 1)$
30 : $P_{30} = (3, 1, 1)$	74 : $P_{76} = (10, 4, 1)$	118 : $P_{124} = (6, 8, 1)$
31 : $P_{31} = (4, 1, 1)$	75 : $P_{77} = (11, 4, 1)$	119 : $P_{125} = (7, 8, 1)$
32 : $P_{32} = (5, 1, 1)$	76 : $P_{78} = (12, 4, 1)$	120 : $P_{126} = (8, 8, 1)$
33 : $P_{33} = (6, 1, 1)$	77 : $P_{79} = (0, 5, 1)$	121 : $P_{127} = (9, 8, 1)$
34 : $P_{34} = (7, 1, 1)$	78 : $P_{81} = (2, 5, 1)$	122 : $P_{128} = (10, 8, 1)$
35 : $P_{36} = (9, 1, 1)$	79 : $P_{82} = (3, 5, 1)$	123 : $P_{129} = (11, 8, 1)$
36 : $P_{37} = (10, 1, 1)$	80 : $P_{83} = (4, 5, 1)$	124 : $P_{130} = (12, 8, 1)$
37 : $P_{38} = (11, 1, 1)$	81 : $P_{85} = (6, 5, 1)$	125 : $P_{131} = (0, 9, 1)$
38 : $P_{40} = (0, 2, 1)$	82 : $P_{86} = (7, 5, 1)$	126 : $P_{132} = (1, 9, 1)$
39 : $P_{41} = (1, 2, 1)$	83 : $P_{87} = (8, 5, 1)$	127 : $P_{133} = (2, 9, 1)$
40 : $P_{42} = (2, 2, 1)$	84 : $P_{88} = (9, 5, 1)$	128 : $P_{134} = (3, 9, 1)$
41 : $P_{43} = (3, 2, 1)$	85 : $P_{89} = (10, 5, 1)$	129 : $P_{135} = (4, 9, 1)$
42 : $P_{44} = (4, 2, 1)$	86 : $P_{90} = (11, 5, 1)$	130 : $P_{136} = (5, 9, 1)$
43 : $P_{45} = (5, 2, 1)$	87 : $P_{91} = (12, 5, 1)$	131 : $P_{137} = (6, 9, 1)$

132 : $P_{138} = (7, 9, 1)$	147 : $P_{153} = (9, 10, 1)$	162 : $P_{168} = (11, 11, 1)$
133 : $P_{139} = (8, 9, 1)$	148 : $P_{154} = (10, 10, 1)$	163 : $P_{169} = (12, 11, 1)$
134 : $P_{140} = (9, 9, 1)$	149 : $P_{155} = (11, 10, 1)$	164 : $P_{170} = (0, 12, 1)$
135 : $P_{141} = (10, 9, 1)$	150 : $P_{156} = (12, 10, 1)$	165 : $P_{171} = (1, 12, 1)$
136 : $P_{142} = (11, 9, 1)$	151 : $P_{157} = (0, 11, 1)$	166 : $P_{172} = (2, 12, 1)$
137 : $P_{143} = (12, 9, 1)$	152 : $P_{158} = (1, 11, 1)$	167 : $P_{173} = (3, 12, 1)$
138 : $P_{144} = (0, 10, 1)$	153 : $P_{159} = (2, 11, 1)$	168 : $P_{174} = (4, 12, 1)$
139 : $P_{145} = (1, 10, 1)$	154 : $P_{160} = (3, 11, 1)$	169 : $P_{175} = (5, 12, 1)$
140 : $P_{146} = (2, 10, 1)$	155 : $P_{161} = (4, 11, 1)$	170 : $P_{177} = (7, 12, 1)$
141 : $P_{147} = (3, 10, 1)$	156 : $P_{162} = (5, 11, 1)$	171 : $P_{179} = (9, 12, 1)$
142 : $P_{148} = (4, 10, 1)$	157 : $P_{163} = (6, 11, 1)$	172 : $P_{180} = (10, 12, 1)$
143 : $P_{149} = (5, 10, 1)$	158 : $P_{164} = (7, 11, 1)$	173 : $P_{181} = (11, 12, 1)$
144 : $P_{150} = (6, 10, 1)$	159 : $P_{165} = (8, 11, 1)$	174 : $P_{182} = (12, 12, 1)$
145 : $P_{151} = (7, 10, 1)$	160 : $P_{166} = (9, 11, 1)$	
146 : $P_{152} = (8, 10, 1)$	161 : $P_{167} = (10, 11, 1)$	

(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 36, 37, 38, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 81, 82, 83, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 111, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 177, 179, 180, 181, 182)

Duals of Bitangents

dual of bitangents:

- 0 : $P_{118} = (0, 8, 1)$
- 1 : $P_{18} = (3, 0, 1)$
- 2 : $P_0 = (1, 0, 0)$
- 3 : $P_{76} = (10, 4, 1)$
- 4 : $P_{146} = (2, 10, 1)$
- 5 : $P_{43} = (3, 2, 1)$
- 6 : $P_7 = (4, 1, 0)$
- 7 : $P_{28} = (0, 1, 1)$
- 8 : $P_{123} = (5, 8, 1)$
- 9 : $P_{111} = (6, 7, 1)$
- 10 : $P_{151} = (7, 10, 1)$
- 11 : $P_{54} = (1, 3, 1)$
- 12 : $P_{105} = (0, 7, 1)$
- 13 : $P_{158} = (1, 11, 1)$
- 14 : $P_{33} = (6, 1, 1)$

$$\begin{aligned}
15 : P_{94} &= (2, 6, 1) \\
16 : P_{136} &= (5, 9, 1) \\
17 : P_{69} &= (3, 4, 1) \\
18 : P_{74} &= (8, 4, 1) \\
19 : P_{138} &= (7, 9, 1) \\
20 : P_{11} &= (8, 1, 0) \\
21 : P_{25} &= (10, 0, 1) \\
22 : P_{102} &= (10, 6, 1) \\
23 : P_{171} &= (1, 12, 1) \\
24 : P_{58} &= (5, 3, 1) \\
25 : P_{87} &= (8, 5, 1) \\
26 : P_{48} &= (8, 2, 1) \\
27 : P_{157} &= (0, 11, 1)
\end{aligned}$$

Dimension of the vanishing ideal is 0

The lines and their points of contact are:

$$\begin{aligned}
a_1 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 5 \end{bmatrix}_5 \\
a_2 &= \begin{bmatrix} 1 & 0 & 10 \\ 0 & 1 & 0 \end{bmatrix}_{140} \\
a_3 &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{182} \\
a_4 &= \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 9 \end{bmatrix}_{51} \\
a_5 &= \begin{bmatrix} 1 & 0 & 11 \\ 0 & 1 & 3 \end{bmatrix}_{157} \\
a_6 &= \begin{bmatrix} 1 & 0 & 10 \\ 0 & 1 & 11 \end{bmatrix}_{151} \quad P_{39} = \mathbf{P}(12, 1, 1) \ 2\times, P_{84} = \mathbf{P}(5, 5, 1) \ 2\times \\
b_1 &= \begin{bmatrix} 1 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{139} \\
b_2 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 12 \end{bmatrix}_{12} \quad P_{176} = \mathbf{P}(6, 12, 1) \ 2\times, P_{178} = \mathbf{P}(8, 12, 1) \ 2\times \\
b_3 &= \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 5 \end{bmatrix}_{117} \\
b_4 &= \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 6 \end{bmatrix}_{104} \\
b_5 &= \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 3 \end{bmatrix}_{87} \\
b_6 &= \begin{bmatrix} 1 & 0 & 12 \\ 0 & 1 & 10 \end{bmatrix}_{178} \\
c_{12} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 6 \end{bmatrix}_6
\end{aligned}$$

$$\begin{aligned}
c_{13} &= \begin{bmatrix} 1 & 0 & 12 \\ 0 & 1 & 2 \end{bmatrix}_{170} \\
c_{14} &= \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 12 \end{bmatrix}_{110} \\
c_{15} &= \begin{bmatrix} 1 & 0 & 11 \\ 0 & 1 & 7 \end{bmatrix}_{161} \\
c_{16} &= \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 4 \end{bmatrix}_{116} \\
c_{23} &= \begin{bmatrix} 1 & 0 & 10 \\ 0 & 1 & 9 \end{bmatrix}_{149} \\
c_{24} &= \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 9 \end{bmatrix}_{79} \\
c_{25} &= \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 4 \end{bmatrix}_{88} \\
c_{26} &= \begin{bmatrix} 1 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{83} \\
c_{34} &= \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \end{bmatrix}_{42} \\
c_{35} &= \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 7 \end{bmatrix}_{49} \\
c_{36} &= \begin{bmatrix} 1 & 0 & 12 \\ 0 & 1 & 1 \end{bmatrix}_{169} \\
c_{45} &= \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 10 \end{bmatrix}_{122} \\
c_{46} &= \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 8 \end{bmatrix}_{78} \\
c_{56} &= \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 11 \end{bmatrix}_{81} \\
d &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \end{bmatrix}_2
\end{aligned}$$

$$P_{35} = \mathbf{P}(8, 1, 1) \ 2 \times, P_{80} = \mathbf{P}(1, 5, 1) \ 2 \times$$

Rank of lines: (5, 140, 182, 51, 157, 151, 139, 12, 117, 104, 87, 178, 6, 170, 110, 161, 116, 149, 79, 88, 83, 42, 49, 169, 122, 78, 81, 2)

Line type: $2^4, 0^{24}$

4	2	27, 5, 20, 7
24	0	13, 25, 1, 0, 2, 26, 3, 8, 4, 10, 11, 12, 6, 14, 15, 16, 17, 18, 19, 9, 21, 22, 23, 24

point types: 1^8

8	1	1, 2, 3, 4, 5, 6, 7, 0
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point types for points off the curve: $4^9, 3^{100}, 2^{24}, 0^{42}$

9	4	107, 105, 51, 11, 5, 171, 169, 4, 0
100	3	62, 123, 30, 60, 120, 119, 117, 116, 115, 113, 55, 27, 13, 6, 109, 53, 170, 106, 103, 85, 84, 101, 24, 48, 96, 47, 23, 2, 168, 93, 97, 92, 45, 89, 44, 102, 88, 104, 87, 173, 42, 108, 9, 110, 111, 112, 41, 114, 167, 20, 1, 82, 164, 163, 80, 122, 160, 124, 125, 126, 159, 39, 19, 174, 78, 156, 38, 76, 152, 151, 37, 74, 147, 73, 146, 144, 71, 35, 70, 140, 138, 68, 136, 150, 67, 33, 153, 154, 132, 130, 157, 129, 64, 31, 161, 162, 3, 172, 165, 166
24	2	141, 59, 155, 99, 95, 90, 26, 21, 28, 118, 56, 61, 121, 54, 77, 142, 17, 143, 34, 135, 158, 134, 133, 81
42	0	94, 52, 137, 91, 46, 10, 63, 145, 57, 22, 148, 29, 69, 25, 83, 40, 72, 8, 16, 58, 12, 100, 49, 127, 43, 14, 131, 139, 149, 50, 79, 65, 36, 32, 66, 75, 86, 18, 15, 98, 7, 128

Lines on points off the curve:

- Off point $0 = P_0 = (1, 0, 0)$ lies on 4 bisecants : $\{ 0, 7, 12, 27 \}$
- Off point $1 = P_1 = (0, 1, 0)$ lies on 3 bisecants : $\{ 1, 2, 21 \}$
- Off point $2 = P_2 = (0, 0, 1)$ lies on 3 bisecants : $\{ 2, 6, 20 \}$
- Off point $3 = P_3 = (1, 1, 1)$ lies on 3 bisecants : $\{ 4, 13, 18 \}$
- Off point $4 = P_4 = (1, 1, 0)$ lies on 4 bisecants : $\{ 8, 9, 23, 25 \}$
- Off point $5 = P_5 = (2, 1, 0)$ lies on 4 bisecants : $\{ 13, 14, 22, 24 \}$
- Off point $6 = P_6 = (3, 1, 0)$ lies on 3 bisecants : $\{ 6, 17, 26 \}$
- Off point $7 = P_7 = (4, 1, 0)$ lies on 0 bisecants : $\{ \}$
- Off point $8 = P_8 = (5, 1, 0)$ lies on 0 bisecants : $\{ \}$
- Off point $9 = P_9 = (6, 1, 0)$ lies on 3 bisecants : $\{ 10, 16, 18 \}$
- Off point $10 = P_{10} = (7, 1, 0)$ lies on 0 bisecants : $\{ \}$
- Off point $11 = P_{11} = (8, 1, 0)$ lies on 4 bisecants : $\{ 4, 5, 19, 20 \}$
- Off point $12 = P_{12} = (9, 1, 0)$ lies on 0 bisecants : $\{ \}$
- Off point $13 = P_{13} = (10, 1, 0)$ lies on 3 bisecants : $\{ 3, 11, 15 \}$
- Off point $14 = P_{14} = (11, 1, 0)$ lies on 0 bisecants : $\{ \}$
- Off point $15 = P_{15} = (12, 1, 0)$ lies on 0 bisecants : $\{ \}$
- Off point $16 = P_{16} = (1, 0, 1)$ lies on 0 bisecants : $\{ \}$
- Off point $17 = P_{17} = (2, 0, 1)$ lies on 2 bisecants : $\{ 9, 14 \}$
- Off point $18 = P_{18} = (3, 0, 1)$ lies on 0 bisecants : $\{ \}$
- Off point $19 = P_{19} = (4, 0, 1)$ lies on 3 bisecants : $\{ 1, 5, 17 \}$
- Off point $20 = P_{20} = (5, 0, 1)$ lies on 3 bisecants : $\{ 8, 16, 24 \}$
- Off point $21 = P_{21} = (6, 0, 1)$ lies on 2 bisecants : $\{ 4, 15 \}$
- Off point $22 = P_{22} = (7, 0, 1)$ lies on 0 bisecants : $\{ \}$

Off point 23 = $P_{23} = (8, 0, 1)$ lies on 3 bisecants : { 18, 25, 26 }
 Off point 24 = $P_{24} = (9, 0, 1)$ lies on 3 bisecants : { 3, 21, 22 }
 Off point 25 = $P_{25} = (10, 0, 1)$ lies on 0 bisecants : { }
 Off point 26 = $P_{26} = (11, 0, 1)$ lies on 2 bisecants : { 10, 19 }
 Off point 27 = $P_{27} = (12, 0, 1)$ lies on 3 bisecants : { 11, 13, 23 }
 Off point 28 = $P_{28} = (0, 1, 1)$ lies on 2 bisecants : { 2, 23 }
 Off point 29 = $P_{29} = (2, 1, 1)$ lies on 0 bisecants : { }
 Off point 30 = $P_{30} = (3, 1, 1)$ lies on 3 bisecants : { 6, 9, 15 }
 Off point 31 = $P_{31} = (4, 1, 1)$ lies on 3 bisecants : { 1, 10, 14 }
 Off point 32 = $P_{32} = (5, 1, 1)$ lies on 0 bisecants : { }
 Off point 33 = $P_{33} = (6, 1, 1)$ lies on 3 bisecants : { 3, 8, 19 }
 Off point 34 = $P_{34} = (7, 1, 1)$ lies on 2 bisecants : { 17, 24 }
 Off point 35 = $P_{36} = (9, 1, 1)$ lies on 3 bisecants : { 11, 21, 25 }
 Off point 36 = $P_{37} = (10, 1, 1)$ lies on 0 bisecants : { }
 Off point 37 = $P_{38} = (11, 1, 1)$ lies on 3 bisecants : { 16, 22, 26 }
 Off point 38 = $P_{40} = (0, 2, 1)$ lies on 3 bisecants : { 2, 15, 22 }
 Off point 39 = $P_{41} = (1, 2, 1)$ lies on 3 bisecants : { 19, 23, 26 }
 Off point 40 = $P_{42} = (2, 2, 1)$ lies on 0 bisecants : { }
 Off point 41 = $P_{43} = (3, 2, 1)$ lies on 3 bisecants : { 3, 13, 20 }
 Off point 42 = $P_{44} = (4, 2, 1)$ lies on 3 bisecants : { 1, 9, 16 }
 Off point 43 = $P_{45} = (5, 2, 1)$ lies on 0 bisecants : { }
 Off point 44 = $P_{46} = (6, 2, 1)$ lies on 3 bisecants : { 6, 11, 14 }
 Off point 45 = $P_{47} = (7, 2, 1)$ lies on 3 bisecants : { 5, 8, 18 }
 Off point 46 = $P_{48} = (8, 2, 1)$ lies on 0 bisecants : { }
 Off point 47 = $P_{49} = (9, 2, 1)$ lies on 3 bisecants : { 4, 21, 24 }
 Off point 48 = $P_{50} = (10, 2, 1)$ lies on 3 bisecants : { 10, 17, 25 }
 Off point 49 = $P_{51} = (11, 2, 1)$ lies on 0 bisecants : { }
 Off point 50 = $P_{52} = (12, 2, 1)$ lies on 0 bisecants : { }
 Off point 51 = $P_{53} = (0, 3, 1)$ lies on 4 bisecants : { 2, 3, 17, 18 }
 Off point 52 = $P_{54} = (1, 3, 1)$ lies on 0 bisecants : { }
 Off point 53 = $P_{55} = (2, 3, 1)$ lies on 3 bisecants : { 5, 22, 23 }
 Off point 54 = $P_{56} = (3, 3, 1)$ lies on 2 bisecants : { 10, 11 }
 Off point 55 = $P_{57} = (4, 3, 1)$ lies on 3 bisecants : { 1, 4, 26 }
 Off point 56 = $P_{58} = (5, 3, 1)$ lies on 2 bisecants : { 9, 13 }
 Off point 57 = $P_{59} = (6, 3, 1)$ lies on 0 bisecants : { }
 Off point 58 = $P_{60} = (7, 3, 1)$ lies on 0 bisecants : { }
 Off point 59 = $P_{61} = (8, 3, 1)$ lies on 2 bisecants : { 8, 14 }
 Off point 60 = $P_{62} = (9, 3, 1)$ lies on 3 bisecants : { 6, 19, 21 }
 Off point 61 = $P_{63} = (10, 3, 1)$ lies on 2 bisecants : { 15, 16 }
 Off point 62 = $P_{64} = (11, 3, 1)$ lies on 3 bisecants : { 20, 24, 25 }
 Off point 63 = $P_{65} = (12, 3, 1)$ lies on 0 bisecants : { }
 Off point 64 = $P_{66} = (0, 4, 1)$ lies on 3 bisecants : { 2, 11, 24 }
 Off point 65 = $P_{67} = (1, 4, 1)$ lies on 0 bisecants : { }
 Off point 66 = $P_{68} = (2, 4, 1)$ lies on 0 bisecants : { }
 Off point 67 = $P_{69} = (3, 4, 1)$ lies on 3 bisecants : { 16, 17, 23 }
 Off point 68 = $P_{70} = (4, 4, 1)$ lies on 3 bisecants : { 1, 19, 22 }

Off point 69 = $P_{71} = (5, 4, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 70 = $P_{72} = (6, 4, 1)$ lies on 3 bisecants : $\{ 9, 18, 20 \}$
 Off point 71 = $P_{73} = (7, 4, 1)$ lies on 3 bisecants : $\{ 13, 15, 26 \}$
 Off point 72 = $P_{74} = (8, 4, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 73 = $P_{75} = (9, 4, 1)$ lies on 3 bisecants : $\{ 8, 10, 21 \}$
 Off point 74 = $P_{76} = (10, 4, 1)$ lies on 3 bisecants : $\{ 3, 5, 14 \}$
 Off point 75 = $P_{77} = (11, 4, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 76 = $P_{78} = (12, 4, 1)$ lies on 3 bisecants : $\{ 4, 6, 25 \}$
 Off point 77 = $P_{79} = (0, 5, 1)$ lies on 2 bisecants : $\{ 2, 25 \}$
 Off point 78 = $P_{81} = (2, 5, 1)$ lies on 3 bisecants : $\{ 6, 10, 24 \}$
 Off point 79 = $P_{82} = (3, 5, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 80 = $P_{83} = (4, 5, 1)$ lies on 3 bisecants : $\{ 1, 15, 23 \}$
 Off point 81 = $P_{85} = (6, 5, 1)$ lies on 2 bisecants : $\{ 17, 22 \}$
 Off point 82 = $P_{86} = (7, 5, 1)$ lies on 3 bisecants : $\{ 3, 4, 9 \}$
 Off point 83 = $P_{87} = (8, 5, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 84 = $P_{88} = (9, 5, 1)$ lies on 3 bisecants : $\{ 13, 16, 21 \}$
 Off point 85 = $P_{89} = (10, 5, 1)$ lies on 3 bisecants : $\{ 8, 11, 26 \}$
 Off point 86 = $P_{90} = (11, 5, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 87 = $P_{91} = (12, 5, 1)$ lies on 3 bisecants : $\{ 14, 18, 19 \}$
 Off point 88 = $P_{92} = (0, 6, 1)$ lies on 3 bisecants : $\{ 2, 5, 26 \}$
 Off point 89 = $P_{93} = (1, 6, 1)$ lies on 3 bisecants : $\{ 14, 15, 25 \}$
 Off point 90 = $P_{94} = (2, 6, 1)$ lies on 2 bisecants : $\{ 4, 16 \}$
 Off point 91 = $P_{95} = (3, 6, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 92 = $P_{96} = (4, 6, 1)$ lies on 3 bisecants : $\{ 1, 3, 24 \}$
 Off point 93 = $P_{97} = (5, 6, 1)$ lies on 3 bisecants : $\{ 6, 18, 23 \}$
 Off point 94 = $P_{98} = (6, 6, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 95 = $P_{99} = (7, 6, 1)$ lies on 2 bisecants : $\{ 11, 19 \}$
 Off point 96 = $P_{100} = (8, 6, 1)$ lies on 3 bisecants : $\{ 9, 10, 22 \}$
 Off point 97 = $P_{101} = (9, 6, 1)$ lies on 3 bisecants : $\{ 17, 20, 21 \}$
 Off point 98 = $P_{102} = (10, 6, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 99 = $P_{103} = (11, 6, 1)$ lies on 2 bisecants : $\{ 8, 13 \}$
 Off point 100 = $P_{104} = (12, 6, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 101 = $P_{105} = (0, 7, 1)$ lies on 3 bisecants : $\{ 2, 13, 27 \}$
 Off point 102 = $P_{106} = (1, 7, 1)$ lies on 3 bisecants : $\{ 3, 10, 27 \}$
 Off point 103 = $P_{107} = (2, 7, 1)$ lies on 3 bisecants : $\{ 19, 25, 27 \}$
 Off point 104 = $P_{108} = (3, 7, 1)$ lies on 3 bisecants : $\{ 14, 26, 27 \}$
 Off point 105 = $P_{109} = (4, 7, 1)$ lies on 4 bisecants : $\{ 1, 11, 20, 27 \}$
 Off point 106 = $P_{111} = (6, 7, 1)$ lies on 3 bisecants : $\{ 23, 24, 27 \}$
 Off point 107 = $P_{113} = (8, 7, 1)$ lies on 4 bisecants : $\{ 5, 6, 16, 27 \}$
 Off point 108 = $P_{114} = (9, 7, 1)$ lies on 3 bisecants : $\{ 9, 21, 27 \}$
 Off point 109 = $P_{115} = (10, 7, 1)$ lies on 3 bisecants : $\{ 4, 22, 27 \}$
 Off point 110 = $P_{116} = (11, 7, 1)$ lies on 3 bisecants : $\{ 15, 18, 27 \}$
 Off point 111 = $P_{117} = (12, 7, 1)$ lies on 3 bisecants : $\{ 8, 17, 27 \}$
 Off point 112 = $P_{118} = (0, 8, 1)$ lies on 3 bisecants : $\{ 0, 2, 8 \}$
 Off point 113 = $P_{119} = (1, 8, 1)$ lies on 3 bisecants : $\{ 0, 11, 16 \}$
 Off point 114 = $P_{120} = (2, 8, 1)$ lies on 3 bisecants : $\{ 0, 13, 17 \}$

Off point 115 = $P_{121} = (3, 8, 1)$ lies on 3 bisecants : $\{ 0, 5, 25 \}$
 Off point 116 = $P_{122} = (4, 8, 1)$ lies on 3 bisecants : $\{ 0, 1, 18 \}$
 Off point 117 = $P_{123} = (5, 8, 1)$ lies on 3 bisecants : $\{ 0, 4, 14 \}$
 Off point 118 = $P_{124} = (6, 8, 1)$ lies on 2 bisecants : $\{ 0, 26 \}$
 Off point 119 = $P_{125} = (7, 8, 1)$ lies on 3 bisecants : $\{ 0, 10, 23 \}$
 Off point 120 = $P_{126} = (8, 8, 1)$ lies on 3 bisecants : $\{ 0, 15, 24 \}$
 Off point 121 = $P_{127} = (9, 8, 1)$ lies on 2 bisecants : $\{ 0, 21 \}$
 Off point 122 = $P_{128} = (10, 8, 1)$ lies on 3 bisecants : $\{ 0, 9, 19 \}$
 Off point 123 = $P_{129} = (11, 8, 1)$ lies on 3 bisecants : $\{ 0, 3, 6 \}$
 Off point 124 = $P_{130} = (12, 8, 1)$ lies on 3 bisecants : $\{ 0, 20, 22 \}$
 Off point 125 = $P_{131} = (0, 9, 1)$ lies on 3 bisecants : $\{ 2, 4, 10 \}$
 Off point 126 = $P_{132} = (1, 9, 1)$ lies on 3 bisecants : $\{ 6, 8, 22 \}$
 Off point 127 = $P_{133} = (2, 9, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 128 = $P_{134} = (3, 9, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 129 = $P_{135} = (4, 9, 1)$ lies on 3 bisecants : $\{ 1, 13, 25 \}$
 Off point 130 = $P_{136} = (5, 9, 1)$ lies on 3 bisecants : $\{ 15, 17, 19 \}$
 Off point 131 = $P_{137} = (6, 9, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 132 = $P_{138} = (7, 9, 1)$ lies on 3 bisecants : $\{ 14, 16, 20 \}$
 Off point 133 = $P_{139} = (8, 9, 1)$ lies on 2 bisecants : $\{ 3, 23 \}$
 Off point 134 = $P_{140} = (9, 9, 1)$ lies on 2 bisecants : $\{ 21, 26 \}$
 Off point 135 = $P_{141} = (10, 9, 1)$ lies on 2 bisecants : $\{ 18, 24 \}$
 Off point 136 = $P_{142} = (11, 9, 1)$ lies on 3 bisecants : $\{ 5, 9, 11 \}$
 Off point 137 = $P_{143} = (12, 9, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 138 = $P_{144} = (0, 10, 1)$ lies on 3 bisecants : $\{ 2, 16, 19 \}$
 Off point 139 = $P_{145} = (1, 10, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 140 = $P_{146} = (2, 10, 1)$ lies on 3 bisecants : $\{ 8, 15, 20 \}$
 Off point 141 = $P_{147} = (3, 10, 1)$ lies on 2 bisecants : $\{ 18, 22 \}$
 Off point 142 = $P_{148} = (4, 10, 1)$ lies on 2 bisecants : $\{ 1, 6 \}$
 Off point 143 = $P_{149} = (5, 10, 1)$ lies on 2 bisecants : $\{ 3, 25 \}$
 Off point 144 = $P_{150} = (6, 10, 1)$ lies on 3 bisecants : $\{ 5, 10, 13 \}$
 Off point 145 = $P_{151} = (7, 10, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 146 = $P_{152} = (8, 10, 1)$ lies on 3 bisecants : $\{ 4, 11, 17 \}$
 Off point 147 = $P_{153} = (9, 10, 1)$ lies on 3 bisecants : $\{ 14, 21, 23 \}$
 Off point 148 = $P_{154} = (10, 10, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 149 = $P_{155} = (11, 10, 1)$ lies on 0 bisecants : $\{ \}$
 Off point 150 = $P_{156} = (12, 10, 1)$ lies on 3 bisecants : $\{ 9, 24, 26 \}$
 Off point 151 = $P_{157} = (0, 11, 1)$ lies on 3 bisecants : $\{ 2, 9, 12 \}$
 Off point 152 = $P_{158} = (1, 11, 1)$ lies on 3 bisecants : $\{ 5, 12, 24 \}$
 Off point 153 = $P_{159} = (2, 11, 1)$ lies on 3 bisecants : $\{ 3, 12, 26 \}$
 Off point 154 = $P_{160} = (3, 11, 1)$ lies on 3 bisecants : $\{ 4, 8, 12 \}$
 Off point 155 = $P_{161} = (4, 11, 1)$ lies on 2 bisecants : $\{ 1, 12 \}$
 Off point 156 = $P_{162} = (5, 11, 1)$ lies on 3 bisecants : $\{ 11, 12, 22 \}$
 Off point 157 = $P_{163} = (6, 11, 1)$ lies on 3 bisecants : $\{ 12, 16, 25 \}$
 Off point 158 = $P_{164} = (7, 11, 1)$ lies on 2 bisecants : $\{ 6, 12 \}$
 Off point 159 = $P_{165} = (8, 11, 1)$ lies on 3 bisecants : $\{ 12, 13, 19 \}$
 Off point 160 = $P_{166} = (9, 11, 1)$ lies on 3 bisecants : $\{ 12, 18, 21 \}$

Off point 161 = $P_{167} = (10, 11, 1)$ lies on 3 bisecants : $\{ 12, 20, 23 \}$
 Off point 162 = $P_{168} = (11, 11, 1)$ lies on 3 bisecants : $\{ 12, 14, 17 \}$
 Off point 163 = $P_{169} = (12, 11, 1)$ lies on 3 bisecants : $\{ 10, 12, 15 \}$
 Off point 164 = $P_{170} = (0, 12, 1)$ lies on 3 bisecants : $\{ 2, 7, 14 \}$
 Off point 165 = $P_{171} = (1, 12, 1)$ lies on 3 bisecants : $\{ 7, 9, 17 \}$
 Off point 166 = $P_{172} = (2, 12, 1)$ lies on 3 bisecants : $\{ 7, 11, 18 \}$
 Off point 167 = $P_{173} = (3, 12, 1)$ lies on 3 bisecants : $\{ 7, 19, 24 \}$
 Off point 168 = $P_{174} = (4, 12, 1)$ lies on 3 bisecants : $\{ 1, 7, 8 \}$
 Off point 169 = $P_{175} = (5, 12, 1)$ lies on 4 bisecants : $\{ 7, 10, 20, 26 \}$
 Off point 170 = $P_{177} = (7, 12, 1)$ lies on 3 bisecants : $\{ 7, 22, 25 \}$
 Off point 171 = $P_{179} = (9, 12, 1)$ lies on 4 bisecants : $\{ 5, 7, 15, 21 \}$
 Off point 172 = $P_{180} = (10, 12, 1)$ lies on 3 bisecants : $\{ 6, 7, 13 \}$
 Off point 173 = $P_{181} = (11, 12, 1)$ lies on 3 bisecants : $\{ 4, 7, 23 \}$
 Off point 174 = $P_{182} = (12, 12, 1)$ lies on 3 bisecants : $\{ 3, 7, 16 \}$

The gradient

The gradient of the quartic curve is :

$$\begin{aligned}
 & 3X_0^3 + 7X_1^3 + 2X_2^3 + 6X_0^2X_1 + 8X_0^2X_2 + 11X_0X_1^2 + 2X_1^2X_2 + 10X_0X_2^2 \\
 & \quad + 4X_1X_2^2 + 12X_0X_1X_2 \\
 & (3, 7, 2, 6, 8, 11, 2, 10, 4, 12) \\
 & 2X_0^3 + 3X_1^3 + 8X_2^3 + 11X_0^2X_1 + 6X_0^2X_2 + 8X_0X_1^2 + 12X_1^2X_2 + 4X_0X_2^2 \\
 & \quad + 8X_1X_2^2 + 4X_0X_1X_2 \\
 & (2, 3, 8, 11, 6, 8, 12, 4, 8, 4) \\
 & 7X_0^3 + 4X_1^3 + 12X_2^3 + 6X_0^2X_1 + 10X_0^2X_2 + 2X_0X_1^2 + 8X_1^2X_2 + 6X_0X_2^2 \\
 & \quad + 11X_1X_2^2 + 8X_0X_1X_2 \\
 & (7, 4, 12, 6, 10, 2, 8, 6, 11, 8)
 \end{aligned}$$